

1. The top-performing departments by calculating the total sales volume generated within each product_category	
product_category	total_sales
Phones	3321700
Laptops	3317450
Other Electronics	3169550
Chargers & Cables	815000
Cameras	814400
TV & Display	696950
Power & Batteries	530400
Storage	440900
Printers & Scanners	290150
Networking	265950
Speakers	224350
Headphones	216750
Gaming	101600
Smart Home	88150
Wearables	54050

2. Top 10 best-selling products that maintain a high customer satisfaction product_rating (above 4.5)		
product_title	purchased_last_month	product_rating
Amazon Basics Wood-Cased #2 Pencils, Pre-sharpened, HB Lead Bulk Box, 150 Count, Yellow	100000	4.8
Texas Instruments TI-84 Plus CE Color Graphing Calculator, Black	100000	4.6
Amazon Basics 48-Pack AA Alkaline High-Performance Batteries, 1.5 Volt, 10-Year Shelf Life	100000	4.7
Texas Instruments TI-30XIIS Scientific Calculator, Black with Blue Accents	100000	4.7
Amazon Basics AAA Alkaline High-Performance Batteries, 1.5 Volt, 10-Year Shelf Life, 36 Count (Pack of 1)	100000	4.7
Amazon Basics Clear Thermal Laminating Plastic Paper Laminator Sheets, 9 x 11.5-Inch, 200-Pack, 3mil	100000	4.8
Amazon Basics Multipurpose Copy Printer Paper, 20 lb, 8.5 x 11 Inches, 8 Reams (4,000 Sheets), 92 Bright, White	80000	4.8
Texas Instruments TI-30XS MultiView Scientific Calculator	70000	4.8
Amazon Basics File Folders with Tabs for Filing, 1/3-Cut Tab, Assorted Positions, 8.5x11 inches, Letter Size, Manila, Pack of 100	70000	4.8
Scotch Thermal Laminating Pouches, for Use with Thermal Laminators, 8.9 x 11.4 Inches, Letter Size Sheets, 100-Pack	70000	4.9

3. Determines whether offering a coupon (has_coupon) significantly increases the average monthly sales volume	
has_coupon	avg_sales
Save 6% with coupon	8000
Save \$2.00 with coupon	6833.333333
Save 7% with coupon	6000

Save \$4.00 with coupon	4750
Save \$1.00 with coupon	2625
Save \$1.50 with coupon	2500
Save \$8.00 with coupon	1750
Save \$18.00 with coupon	1687.5
Save 15% with coupon	1027.638191
Save 8% with coupon	1000
Save 5% with coupon	910.4712042
Save 25% with coupon	833.3333333
Save \$5.00 with coupon	750
Save \$40.00 with coupon	600
Save \$15.00 with coupon	470
Save 10% with coupon	407.3394495
Save \$10.00 with coupon	407.1428571
No Coupon	377.0639259
Save \$12.00 with coupon	366.6666667
Save 20% with coupon	366.6666667
Save 30% with coupon	300
Save \$3.00 with coupon	300
Save 40% with coupon	233.3333333
Save \$50.00 with coupon	200
Save 25% with coupon (some sizes/colors)	200
Save \$90.00 with coupon	200
Save \$25.00 with coupon	196.9230769
Save \$20.00 with coupon	150
Save \$133.00 with coupon	100
Save \$30.00 with coupon	64.28571429
Save \$100.00 with coupon	53.09278351
Save \$13.00 with coupon	50
Save 50% with coupon	50
Save \$69.00 with coupon	50
Save \$11.00 with coupon	3.333333333
Save \$29.00 with coupon	0
Save \$45.00 with coupon	0
Save \$120.00 with coupon	0
Save \$16.00 with coupon	0
Save \$60.00 with coupon	0
Save \$33.00 with coupon	0

4. Which product categories rely the most on heavy price cuts by calculating the average discount_percentage	
product_category	avg_discount
Headphones	13.93818642
Chargers & Cables	12.66784228
Networking	12.30668405
Storage	11.68571958
Printers & Scanners	9.197762557
Phones	8.739455865
Speakers	7.086435496
Laptops	7.050260459
Cameras	4.707289851

Gaming	4.421023766
Power & Batteries	3.537736022
Smart Home	3.357857143
TV & Display	3.307424942
Other Electronics	3.200720471
Wearables	2.446919431

5. Compares paid (is_sponsored) products against organic ones to see which group drives higher average reviews/sales		
is_sponsored	avg_reviews	avg_monthly_sales
Sponsored	3082.393704	449.9611349
Organic	2758.930674	370.8840665

6. Analyzes whether products with the is_best_seller badge generally have a higher or lower average price point	
is_best_seller	avg_price
No Badge	246.0247625
Best Seller	87.92828244
Amazon's	211.0435433
Limited time deal	51.66621795
Save 30%	139.95
Save 12%	579
Save 9%	299
Save 17%	149.99
Save 77%	339.99
Ends in	56.25391753
Save 10%	1579.99
Save 18%	899.99

7. Highlights highly popular products (over 10k reviews) that have a poor product_rating (below 4.0)		
product_title	total_reviews	product_rating
NETGEAR WiFi Range Extender (EX6120) - Add up to 1,500 sq. ft., 25 Devices - AC1200 Dual Band Wireless Signal Booster & Repeater (Up to 1.2 Gbps Speed), Compact Wall Plug Design – Free Expert Help	119219	3.8
NETGEAR WiFi Range Extender (EX6120) - Add up to 1,500 sq. ft., 25 Devices - AC1200 Dual Band Wireless Signal Booster & Repeater (Up to 1.2 Gbps Speed), Compact Wall Plug Design – Free Expert Help	119218	3.8
NETGEAR WiFi Range Extender (EX6120) - Add up to 1,500 sq. ft., 25 Devices - AC1200 Dual Band Wireless Signal Booster & Repeater (Up to 1.2 Gbps Speed), Compact Wall Plug Design – Free Expert Help	119215	3.8

NETGEAR Wi-Fi Range Extender EX6120 - Coverage Up to 1500 Sq Ft and 25 Devices with AC1200 Dual Band Wireless Signal Booster & Repeater (Up to 1200Mbps Speed), and Compact Wall Plug Design, White	119204	3.8
Canon Pixma MG3620 Wireless All-in-One Color Inkjet Printer with Mobile and Tablet Printing, Black	30709	3.9
KODAK Printomatic Full-Color Instant Print Digital Camera - Colorful Photos On Zink 2x3" Sticky-Back Photo Paper - 5MP, Portable, Creative Fun Gift for Birthday, Christmas, Holiday - Blue	27254	3.9
Case-Mate Minis - Cell Phone Stand/Grip - Suction Cup Cell Phone Holder [Removable for Wireless Charging] for iPhones, Android Smartphones, Kindles & More - Twinkle Stardust	19684	3.7
HP DeskJet 2855e Wireless All-in-One Color Inkjet Printer, Scanner, Copier, Best-for-home, 3 months of Instant Ink included, Single-band Wi-Fi connectivity (588S5A)	11214	3.9
JLab Go Work Wireless Headsets with Microphone, 45+ Playtime PC Bluetooth Headset and Multipoint Connect to Laptop Computer and Mobile, Wired or Wireless Headphones (1 Pack)	10355	3.9

8. Best-selling product_title within every single product_category		
product_category	product_title	purchased_last_month
Cameras	Canon PG-275 / CL-276 Genuine Ink Value Pack (2 Cartridges), Compatible with TS3520/3522/4722/3720/3722, TR4720	20000
Chargers & Cables	APC UPS Battery Backup and Surge Protector, 600VA/330 Watts Backup Battery Power Supply, BE600M1 Back-UPS with USB Charger Port	10000
Gaming	Razer Basilisk V3 Customizable Ergonomic Gaming Mouse: Fastest Gaming Mouse Switch - Chroma RGB Lighting - 26K DPI Optical Sensor - 11 Programmable Buttons - HyperScroll Tilt Wheel - Classic Black	6000
Headphones	Apple AirPods 2 with Charging Case - White (Renewed)	10000
Laptops	Amazon Basics 48-Pack AA Alkaline High-Performance Batteries, 1.5 Volt, 10-Year Shelf Life	100000

Networking	TP-Link AC1900 WiFi Range Extender RE550 Dual-Band Wireless Repeater Amplifier w/Gigabit Ethernet Port Up to 2200 Sq. Ft., 32 Devices Internet Signal Booster APP Setup EasyMesh Compatible	10000
Other Electronics	Texas Instruments TI-30XIIS Scientific Calculator, Black with Blue Accents	100000
Phones	LISEN Retractable Car Charger, 69W USB C Car Accessories Adapter for iPhone 16 USB C Charger Fast Charging,Gifts for Women Men, Labor Day Sale, Truck Accessories, for iPhone 16 Pro Max 15 14 13 12	10000
Power & Batteries	Duracell - CopperTop AA Alkaline Batteries - Long Lasting, All-Purpose Double A Battery for Household and Business, 4 Batteries, Power Remotes, Toys, and More, Reliable and Trustworthy	30000
Printers & Scanners	Amazon Basics Multipurpose Copy Printer Paper, 20 lb, 8.5 x 11 Inches, 8 Reams (4,000 Sheets), 92 Bright, White	80000
Smart Home	TP-Link AX1800 WiFi 6 Router V4 (Archer AX21) – Dual Band Wireless Internet, Gigabit, Easy Mesh, Works with Alexa - A Certified for Humans Device, Free Expert Support	10000
Speakers	JBL FLIP 5, Waterproof Portable Bluetooth Speaker, Black, Small	10000
Storage	SanDisk 128GB Extreme PRO SDXC UHS-I Memory Card - C10, U3, V30, 4K UHD, SD Card - SDCSDXXD-128G-GN4IN	10000
TV & Display	Amazon Basics Wide Ruled Lined Writing Note Pads, 8.5 inch x 11.75 inch, White, 12 Count (12 Pack of 50)	30000
Wearables	Fitbit Inspire 3 Health &-Fitness-Tracker with Stress Management, Workout Intensity, Sleep Tracking, 24/7 Heart Rate and more, Midnight Zen/Black One Size (S & L Bands Included)	10000

9. Segment the discounted_price into distinct tiers and calculates the total sales volume for each	
price_tier	total_units_sold
Budget (Under \$25)	6300050
Mid-Range (\$25-\$100)	4915200
Premium (Over \$100)	3132100

10. Items with explosive recent sales (purchased_last_month > 1000) but surprisingly low historical reviews (under 500)		
product_title	purchased_last_month	total_reviews
HP OfficeJet Pro 8125e Wireless All-in-One Color Inkjet Printer, Print, scan, Copy, ADF, Duplex Printing, Home & Office, 3 Months of Instant Ink Included (405T6A)	20000	450
HP OfficeJet Pro 8125e Wireless All-in-One Color Inkjet Printer, Print, scan, Copy, ADF, Duplex Printing, Home & Office, 3 Months of Instant Ink Included (405T6A)	20000	451
LISEN Suction Cup Phone Mount, Vacuum Magnetic Phone Holder, 360° Adjustable for MagSafe Car Holder Foldable Suction Phone Mount for Car, for Garmin GPS/Mirror/Smooth Surface, for iPhone 16 Pro Max 15	9000	412
LISEN Suction Cup Phone Mount, Vacuum Magnetic Phone Holder, 360° Adjustable for MagSafe Car Holder Foldable Suction Phone Mount for Car, for Garmin GPS/Mirror/Smooth Surface, for iPhone 16 Pro Max 15	7000	436
Fujifilm INSTAX Mini Instant Film (White) for Fujifilm Mini 8,9,11,12 Cameras w/Microfiber Cloth by Quality (20 Prints)	6000	391
HP 218X Black High-Yield Toner Cartridge Works with Color Laserjet Pro 3201, MFP 3301 Series W2180X	5000	297
Texas Instruments TI-30XS Multiview Scientific Calculator (Pink)	5000	119
Scotch Thermal Laminator, Extra Wide 13 Inch Input, Ideal for Teachers, Small Offices, or Home (TL1302Z)	5000	378
Fujifilm QuickSnap Flash 400 One Time Use 35mm Camera with Flash, 27 Exposures, 4-Pack	5000	380
Sony BDP-S1700U Blu-ray DVD Player Dolby Digital TrueHD/DTS and DVD upscaling, with Included HDMI Cable, 2025 Model	5000	130
HP 68 Black Ink Cartridge Works Envy 6100e, 6500e Series Eligible for Instant Ink 7FP21TN	5000	413
HP Color Laserjet Pro MFP 3301sdw Wireless All-in-One Color Laser Printer, Scanner, Copier, Best-for-Office (499Q3F)	4000	496
Carhartt 21L Classic Laptop Backpack, Water-Resistant Pack W/ 15" Laptop Sleeve and Organizational Compartments, PFAS Free, Black	4000	281
TI-30XIIS Scientific Calculator, Lavender Limits	4000	408

Acer SD Card Reader USB C, Dual Slots Type C to Memory Card Reader, Aluminum Micro SD Card Camera Adapter for MicroSD, SDHC, SDXC, Compatible with iPhone16/15, MacBook, iPad Pro, Android, Laptop	4000	310
HP 936 Black Ink Cartridge Printers Works with Printer Series: OfficeJet 9120, OfficeJet Pro 9110, 9120, 9130 Series, Wide Format 9730 Eligible for Instant Ink 4S6V2LN	4000	454
HP 138A Black Toner Cartridge Works Laserjet Pro 3001 Series, Laserjet Pro MFP 3101 Series W1380A	4000	457
Acer SD Card Reader USB C, Dual Slots Type C to Memory Card Reader, Aluminum Micro SD Card Camera Adapter for MicroSD, SDHC, SDXC, Compatible with iPhone16/15, MacBook, iPad Pro, Android, Laptop	4000	310
Dell 27 Plus 4K Monitor - S2725QS - 27-inch 4K (3840 x 2160) 120Hz 16:9 Display, IPS Panel, AMD FreeSync Premium, sRGB 99%, Integrated Speakers, 1500:1 Contrast Ratio, Comfortview Plus - Ash White	3000	423
Lenovo Laptop Charger 65W Watt USB Type C(USB-C) AC Power Adapter - ThinkPad T14 T15 E14 E15 X1Carbon X13 2023	3000	283
Ubiquiti Networks UniFi 7 Pro WiFi 7 Access Point US Model PoE+ Adapter not Included (U7-Pro-US)	3000	260
Dell Universal Dock UD22 Black	3000	391
HP 218A Black Toner Cartridge Works with Color Laserjet Pro 3201, MFP 3301 Series W2180A	3000	265
Apple Magic Keyboard with Numeric Keypad: Wireless, Bluetooth, Rechargeable. Works with Mac, iPad, or iPhone; US English - White	3000	63
Baseus Docking Station, Spacemate(Win) 11 in 1, Triple Display, 4K HDMI and DP, 10Gbps USB A/C Ports, Gigabit Ethernet, 100W PD, Audio, Laptop Docking Station for Windows/Dell/HP/Lenovo/Asus/Acer/Mac	3000	360
HP DeskJet 4227e Wireless All-in-One Color Inkjet Printer, Scanner, Copier, Best-for-Home, 3 Months of Instant Ink Trial Included (6W7F0A)	3000	94
Amazfit Helio Strap Fitness Tracker Wristband, 24/7 Activity & Sleep Tracker with Heart Rate, 10 Days Battery, 27 Sports Modes, Strength Training, Hyrox Race, Subscription Free for Android & iPhone	3000	181
Apple Pencil (1st Generation): Device Compatibility Check Required - Pixel-Perfect Precision and Industry-Leading Low Latency, Perfect for Note-Taking, Drawing, and Signing documents.	3000	329
HP 68 Tri-Color Ink Cartridge Works Envy 6100e, 6500e Series Eligible for Instant Ink 7FP20TN	3000	275
Acer SD Card Reader USB C, 3 in 1 Memory Card Reader with Dual Slots & USB 3.0, Micro SD Card Adapter for iPhone 16/15 Compatible with Camera, Android, MacBook Pro/Air, Laptop	3000	143
acer USB C to DisplayPort Cable 6.6 FT (4K@60HZ 2K@165Hz), Type-C to DisplayPort Cord (Sturdy Aluminum, Plug & Play) Thunderbolt 3/4/5 Compatible with MacBook Pro/Air, iPhone 16 Pro/Max, XPS, Surface	3000	348

LISEN Multi Charging Cable, 3 in 1 Charging Cable, Retractable Car Charger with Type C/L/Micro USB Ports for iPhone 16 15 14 Pro Max/Samsung Galaxy S25/PS/Kindle, Multi Chargers for All Devices	3000	472
Canon PIXMA TR160 Wireless Portable Printer, 50-Sheet Paper Tray and 1.44" Display	2000	57
ARRIS (S34) - Cable Modem - Fast DOCSIS 3.1 Multi-Gigabit Cable Modem, for Comcast Xfinity, Cox, Spectrum & More, 2.5 Gbps Max Internet Speeds, 2 Year Warranty - (No Built-in WiFi)	2000	343
Brother Genuine TN830XL 2PK Black High Yield Printer Toner Cartridge 2-Pack – Print up to 3,000 Pages Each(1)	2000	175
HP 218X Yellow High-Yield Toner Cartridge Works with Color Laserjet Pro 3201, MFP 3301 Series W2182X	2000	158
HP 218X Magenta High-Yield Toner Cartridge Works with Color Laserjet Pro 3201, MFP 3301 Series W2183X	2000	125
HP 210A Black Original Laserjet Toner Cartridge	2000	141
TP-Link Archer BE6500 Dual-Band Wi-Fi 7 Router (Archer BE400) Dual 2.5 Gbps Ports USB 3.0 Covers up to 2,400 Sq. ft and 90 Devices Quad-core CPU HomeShield, Private IoT for Network Security	2000	387
SanDisk 128GB Extreme PRO SDXC UHS-II Memory Card - Up to 300MB/s Read and Write speeds, V90, 8K, 4K UHD, SD Card - SDSDXDM-128G-GN4IN	2000	237
Arlo Video Doorbell 2K 2nd Generation & Chime 2 Bundle, 2K Doorbell Camera, 2-Way Audio, Package Detection, Motion Detection and Alerts, Built-in Siren, Night Vision, AVDK4001-100CNS, White	2000	477
HP 923 Cyan/Magenta/Yellow Standard Yield Ink Cartridges, 3/Pack (6C3Y4LN)	2000	304
HP 936 Cyan/Magenta/Yellow Standard Yield Ink Cartridges, 3/Pack (6C3Z3LN)	2000	208
GoPro Dual Battery Charger + 2 Enduro Rechargeable Batteries (HERO13 Black) - Official GoPro Accessory	2000	422
Brother Genuine TN2294PK Standard Yield 4-Pack Printer Toner Cartridges - Black, Cyan, Magenta, Yellow Toner Multipack	2000	490
Brother Genuine TN830 Black Standard Yield Printer Toner Cartridge - Print up to 1,200 Pages(1)	2000	295
Dell Laptop Charger 65W Watt USB Type C AC Power Adapter Include Power Cord for Dell Latitude 3340 3440 3540 5340 5440 5540 7340 7440 7640 9440 2in1	2000	350
HP Original 148X Black Laserjet Toner Cartridge This Cartridge Works Laserjet Pro 4001, MFP 4101 W1480X, Black	2000	273
ARCTIC Liquid Freezer III Pro 360 A-RGB - AIO CPU Cooler, 3 x 120 mm Water Cooling, 38 mm Radiator, PWM Pump, VRM Fan, AMD AM5/AM4, Intel LGA1851/1700 Contact Frame - Black	2000	419
Sceptre New 22-inch Professional Monitor HDMI x2 Build-in Speakers 2025	2000	45
Carhartt 25L Classic Laptop Backpack, Durable Water-Resistant Pack W/ 15" Laptop Sleeve and Multiple Organizational Compartments, PFAS Free, Black	2000	330

Apple 30W USB-C Power Adapter: Fast and Convenient Charging, Type C Wall Charger	2000	401
SanDisk 512GB microSD Express microSD Card - Up to 880MB/s Read, Up to 650MB/s Write, 220MB/s Sustained Write, U3, C10 - SDSQXFN-512G-GN4NN	2000	3
Apple 70W USB-C Power Adapter	2000	220
Apple Thunderbolt 3 (USB-C) to Thunderbolt 2 Adapter	2000	353
HP 210X Black High-Yield Toner Cartridge Works Color Laserjet Pro 4201, Color Laserjet Pro MFP 4301 Series W2100X	2000	191
HP 210A Cyan Toner Cartridge Works Color Laserjet Pro 4201, Color Laserjet Pro MFP 4301 Series W2101A	2000	119
Lenovo WL310 Bluetooth Silent Mouse, Ergonomic, Adjustable DPI, Silent Clicks, Bluetooth 5.0, Portable for Windows, Mac, and Chromebook, Grey	2000	410
Scotch Matte Thermal Laminating Pouches, Ultra Clear with Matte Finish, Letter Size 8.9 in x 11.4 in, 50-Pack	2000	404
Samsung SSD 9100 PRO 2TB, PCIe 5.0x4 M.2 2280, Seq. Read Speeds Up to 14,800MB/s, Best for AI Computing, Gaming, and Heavy Duty Workstations (MZ VAP2T0B/AM)	2000	465
Acer USB C Docking Station Dual Monitor With 2 HDMI, 9-IN-1 Laptop Docking Station with 4K HDMI, USB A&C 3.0, SD/Micro SD, 100W PD Charging, USB C Dock Compatible with Acer/Dell XPS/HP/Mac/Surface etc	2000	328
Dell Optiplex 3060 Desktop Computer Intel i5-8500 (3.2) 32GB DDR4 RAM 1TB SSD Solid State Built in WiFi Bluetooth Windows 11 Professional Home or Office PC (Renewed)	2000	447
HP 210A Yellow Toner Cartridge Works Color Laserjet Pro 4201, Color Laserjet Pro MFP 4301 Series W2102A	2000	114
Baseus Bass BP1 Pro Noise Cancelling Wireless Earbuds, Real-Time Adaptive Noise Cancelling, Adaptive Active EQ, LDAC Hi-Res Sound, 6-Mic AI Clear Calls, 55H Playtime, IP55 Waterproof, Bluetooth 6.0	2000	300
Belkin USB A to USB C Cable 2-Pack, 6.6ft, 15W - Type C Charger, iPhone Charger Cord, Soft, Braided, Flexible for iPhone 16 & 15, Galaxy S25, Pixel 9, iPad Pro, AirPods Pro, Switch & More - Black	2000	240
VELCRO Brand Reusable Cable Ties, 100 Pack 6 Inch Cable Management Ties, Cord Wraps for Electrical Cords, Black	2000	189
BAGSMART Electronics Organizer Travel Case, Cord Storage for Chargers Cables Phones, Compact Tech Accessories Pouch, Pink	2000	292
LISEN 2-Pack USB C to Micro USB Cable 3.3FT Nylon Braided, Micro USB Cable PS4 Controller Charger, 2.4A Charging&480Mbps Data Transmission, for Type C Wall Chargers/PS4/Xbox One/Old Kindle/Android	2000	427
Baseus Bass BC1 Open Ear Earbuds Clip-On Headphones, Open-Ring Comfort with Tri-Curve Snug Clip, Powerful Sound with SuperBass, 4-Mic Clear Calls, IP55 Waterproof, 38H Playtime, Lightweight	2000	445